Ep 4 · LLM Gateway — Cheat Sheet

AI System Design · AI Engineering with Ajit · github.com/byruajit/ai-system-design

INSIDE THE GATEWAY (IN ORDER)

1. **Auth:** virtual key → team
2. **Policy:** data class → allowed models / regions
3. **Budget & rate limit** per team
4. **Router:** task · length · difficulty
5. **Fallback chain:** timeouts + cooldown
6. **Log:** team · model · tokens · cost · latency

MODEL SELECTION (IN ORDER)

Task quality on **your** golden set → data constraints → latency → cost per answer → context window → ops (limits, SLA, regions)

ROUTING PATTERNS

- **Rules:** task type, prompt length
- **Classifier:** cheap difficulty model
- **Cascade:** try small → check → escalate

AVAILABILITY MATH

- One provider at 99.5% ≈ 3.6 h down/month
- Two independent: $1 - 0.005^2 \approx$ **99.9975%**

COST (NORTHWIND, ILLUSTRATIVE)

All frontier **$3,101/mo** → routed 70/30 **$1,031/mo**

SELF-HOST BREAK-EVEN

`tokens/month at which (GPU + ops people) = API cost` Self-host if: data can't leave, sustained high utilisation, fine-tuned open model.

CHANGE SAFETY

Pin versions · shadow on golden set · canary 5→25→100% · instant rollback

NEVER

Keys in code · "latest" aliases · raw PII in logs · fallbacks that break residency

Northwind Bank is fictional. Prices are illustrative tiers — check current provider pricing.